Numerical Study of the Synchronization Transition in the Kuramoto Model: Collective Dynamics and Analogies with Neural Rhythms

Andrés Eduardo Arrieta Lozano*
Universidad de Pamplona, Physics Program, Pamplona, Norte de Santander, Colombia
(Dated: June 20, 2026)

We numerically study the synchronization transition in the Kuramoto model for $N = 500$ coupled phase oscillators, comparing three natural-frequency distributions: Lorentzian, Gaussian, and uniform, all with equal standard deviation $\gamma = 1$. The coupled system of ordinary differential equations is integrated with the fourth-order Runge-Kutta method (RK4) using step size $h = 0.05$. The transition is characterized through the order parameter $r(t)$ as a function of the coupling strength K, identifying the critical coupling K_c for each distribution. The numerical results are validated against the exact analytical solution available for the Lorentzian distribution, $r = \sqrt{1 - K_c/K}$, yielding a root-mean-square error $\varepsilon_{\mathrm{RMS}} = 0.0167$. Spectral analysis via FFT and the study of effective frequencies Ω_i confirm the theoretically predicted coexistence of locked and drifting oscillators. The results are discussed within the framework of a qualitative analogy with neural rhythms, where distinct regimes of r correspond to experimentally observed brain states, including the hypersynchronization characteristic of epileptic seizures.

INTRODUCTION

Spontaneous synchronization of coupled oscillators is one of the most ubiquitous collective phenomena in nature. From fireflies flashing in unison [8] to electrical generators that must operate in phase to stabilize the power grid [9], the emergence of global coordination from local interactions poses fundamental questions in nonlinear dynamical systems.

The model proposed by Kuramoto in 1975 and published in its canonical form in 1984 [1] is the minimal model that captures this transition. It considers N oscillators, each described solely by its phase $\theta_i(t)$ and its natural frequency ω_i, coupled through a sinusoidal mean-field interaction. Despite its simplicity, it exhibits a continuous phase transition, analogous to mean-field ferromagnetism, controlled by a single parameter: the coupling strength K [2]. This elegance is what makes the model an ideal testbed both for dynamical systems theory and for computational physics.

The most fascinating connection of the model is with neuroscience. The human brain contains approximately 86×10^9 neurons forming massively interconnected networks. These neuronal populations exhibit collective oscillations in well-defined frequency bands —δ (0.5–4 Hz), θ (4–8 Hz), α (8–13 Hz), β (13–30 Hz), and γ (30–80 Hz)— associated with distinct cognitive and behavioral states [6]. It is remarkable that a model as simple as Kuramoto's can qualitatively capture the essential physics of these phenomena: computational neuroscience studies have shown that populations of neurons simplified as phase oscillators behave in agreement with the model [4, 5]. Excessive synchronization ($r \to 1$) reproduces the hypersynchronization observed in epileptic seizures, while normal cortical processing operates in the regime of partial incoherence ($r \approx 0$) [4]. This analogy does not imply that the Kuramoto model is a quantitative description of neural dynamics —which would require Hodgkin-Huxley or integrate-and-fire models— but rather that it provides a mean-field approximation to the emergence of collective coordination in complex biological systems [7].

In this work we numerically study the synchronization transition for three natural-frequency distributions: Lorentzian, Gaussian, and uniform. We analyze the order parameter, the effective frequencies, and the power spectrum, validate the numerical results against the analytical prediction for the Lorentzian case, and discuss the results within the framework of the neural analogy.

THEORETICAL BACKGROUND

Kuramoto model. The dynamics of the phase $\theta_i(t)$ of oscillator i is governed by [1]

$$\dot{\theta}_i = \omega_i + \frac{K}{N}\sum_{j=1}^{N}\sin(\theta_j - \theta_i), \quad i = 1, \ldots, N. \qquad (1)$$

The sinusoidal term acts as a phase spring: if oscillator j is ahead of i, the force accelerates i; if it lags behind, it slows i down. The factor $1/N$ normalizes the mean-field coupling so that the total force is $\mathcal{O}(K)$ independently of N [2].

Order parameter. Kuramoto introduced the complex order parameter

$$r(t)\, e^{i\psi(t)} = \frac{1}{N}\sum_{j=1}^{N} e^{i\theta_j(t)}, \qquad (2)$$

where $r \in [0, 1]$ measures the phase coherence and ψ is the mean phase of the ensemble. Geometrically, r is the length of the average vector when the N oscillators are represented as unit vectors on the circle; $r = 0$ corresponds to a uniform phase distribution and $r = 1$ to perfect synchronization.

Mean-field reduction. Multiplying (2) by $e^{-i\theta_i}$ and taking the imaginary part, Eq. (1) can be rewritten in the compact form [3]

$$\dot{\theta}_i = \omega_i + Kr\sin(\psi - \theta_i). \tag{3}$$

This reduction is crucial: each oscillator interacts solely with the mean field (r, ψ), not with the $N-1$ neighbors individually. Computationally, this reduces the complexity per evaluation of the derivative from $\mathcal{O}(N^2)$ to $\mathcal{O}(N)$.

Transition and critical coupling. In the limit $N \to \infty$, for symmetric and unimodal $g(\omega)$, there exists a critical coupling [2]

$$K_c = \frac{2}{\pi\, g(\omega_0)}, \tag{4}$$

such that for $K < K_c$ the only stationary state is $r = 0$, and for $K > K_c$ a synchronized state with $r > 0$ emerges. Near K_c the transition is continuous, with critical behavior $r \sim (K - K_c)^{1/2}$ [3], corresponding to the mean-field Landau exponent $\beta = 1/2$.

Frequency distributions. For the Cauchy-Lorentz distribution $g(\omega) = (\gamma/\pi)[(\omega - \omega_0)^2 + \gamma^2]^{-1}$, the self-consistency equation has a closed-form solution [1]: $r = \sqrt{1 - K_c/K}$ with $K_c = 2\gamma$. For the Gaussian $g(\omega) = (\gamma\sqrt{2\pi})^{-1}\exp[-(\omega - \omega_0)^2/2\gamma^2]$ and the uniform distribution $g(\omega) = (2\sqrt{3}\gamma)^{-1}$ on $|\omega - \omega_0| \le \sqrt{3}\gamma$, no closed-form solution exists; the theoretical critical couplings are $K_c^{(\mathrm{G})} = 2\gamma\sqrt{2\pi}$ and $K_c^{(\mathrm{U})} = 2\pi\gamma/\sqrt{3}$, respectively. All three distributions are parametrized with the same γ to guarantee equal standard deviation and a comparison under equivalent dispersion conditions.

COMPUTATIONAL DETAILS

Simulation setup. We simulated $N = 500$ oscillators with $\omega_0 = 0$ (rotating frame) and $\gamma = 1$. A sweep of $N_K = 40$ values of K in $[0, 6]$ was performed. For each K and distribution the same random seed was used (reproducibility). Initial conditions are $\theta_i(0) \sim \mathcal{U}[0, 2\pi)$.

Frequency sampling. The frequencies ω_i were sampled via: inverse transform for the Lorentzian, $\omega_i = \gamma\tan[\pi(u_i - 1/2)]$ with $u_i \sim \mathcal{U}(0,1)$; NumPy's standard Gaussian generator for the normal distribution; and uniform sampling on $[-\sqrt{3}\gamma, \sqrt{3}\gamma]$.

RK4 integrator. System (1) is integrated with the standard scheme

$$\boldsymbol{\theta}^{n+1} = \boldsymbol{\theta}^n + \frac{h}{6}(\boldsymbol{k}_1 + 2\boldsymbol{k}_2 + 2\boldsymbol{k}_3 + \boldsymbol{k}_4), \tag{5}$$

with $h = 0.05$ satisfying $h \le 0.1/(K + \gamma)$ over the entire range studied. The global error is $\mathcal{O}(h^4)$. The derivative $\boldsymbol{F}(\boldsymbol{\theta})$ is evaluated using form (3): first (r, ψ) in $\mathcal{O}(N)$, then $F_i = \omega_i + Kr\sin(\psi - \theta_i)$ in $\mathcal{O}(N)$, avoiding the $\mathcal{O}(N^2)$ double sum.

Computed observables. For each K a transient $T_{\mathrm{trans}} = 50$ is integrated and discarded, followed by $T_{\mathrm{sim}} = 100$ of measurement. The average order parameter is $\langle r\rangle = N_{\mathrm{sim}}^{-1}\sum_n r(t_n)$ with $N_{\mathrm{sim}} = 2000$ steps. Effective frequencies are estimated as $\Omega_i = \Delta_i\theta/T_{\mathrm{sim}}$, correcting 2π phase jumps. The power spectrum of the fluctuations is obtained by applying `scipy.fft.rfft` to $r(t) - \langle r\rangle$ with frequency resolution $\Delta f = 0.01\,\mathrm{Hz}$. Validation is quantified through the root-mean-square error $\varepsilon_{\mathrm{RMS}} = \{N_>^{-1}\sum_{K_j > K_c}[\langle r\rangle_j - r_{\mathrm{anal}}(K_j)]^2\}^{1/2}$.

RESULTS

Synchronization curve. Fig. 1 shows the average order parameter $\langle r\rangle$ as a function of K for the three distributions, together with the analytical prediction for the Lorentzian case. All three distributions exhibit the expected transition: for small K, $\langle r\rangle \approx 0$, and for $K \gg K_c$ the order parameter approaches 1. The numerically estimated critical coupling differs across distributions: the Gaussian synchronizes at $K_c^{\mathrm{num}} \approx 2.0$, the Lorentzian at $K_c^{\mathrm{num}} \approx 2.0$ (consistent with the theoretical $K_c = 2\gamma = 2$), and the uniform distribution requires a slightly larger coupling. This difference reflects the role of the tails of $g(\omega)$: distributions with heavy tails (Lorentzian) contain oscillators with very extreme frequencies that resist entrainment, raising the effective K needed to sustain $r > 0$.

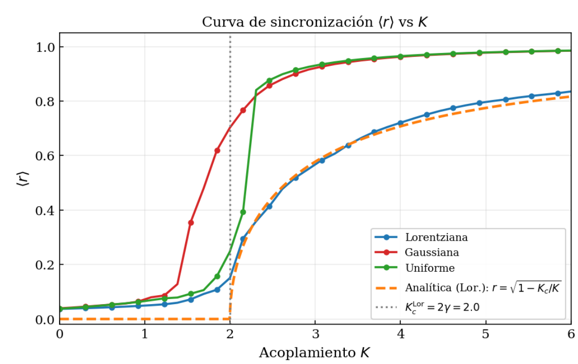

FIG. 1. Average order parameter $\langle r\rangle$ as a function of the coupling K for the Lorentzian, Gaussian, and uniform distributions ($N = 500$, $\gamma = 1$). The dashed orange line corresponds to the exact analytical solution $r = \sqrt{1 - K_c/K}$ for the Lorentzian case. The dotted gray vertical line indicates $K_c = 2\gamma = 2$.

Temporal evolution of the order parameter. Fig. 2 shows $r(t)$ for three representative values of K (Lorentzian distribution). For $K = 0.8 < K_c$, $r(t)$ fluctuates around a small value $\approx 1/\sqrt{N} \approx 0.045$, consistent with the expected finite-size fluctuations [2]; the system does not develop sustained coherence. For $K = 2.0 \approx K_c$, the fluctuations are notably larger, reflecting the system's extreme sensitivity near the transition. For $K = 4.5 > K_c$, $r(t)$ rapidly converges to a steady-state value $\langle r\rangle \approx 0.76$, with small fluctuations of amplitude $\mathcal{O}(1/\sqrt{N})$ around that value.

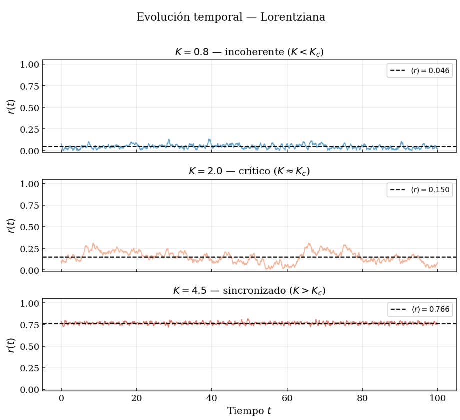

FIG. 2. Temporal evolution of $r(t)$ for the Lorentzian distribution in three regimes: incoherent ($K = 0.8$), critical ($K = 2.0$), and synchronized ($K = 4.5$). The dashed line indicates $\langle r\rangle$.

Phase distribution and effective frequencies. Fig. 3 visually confirms the transition: for $K = 0.8$ the oscillators are approximately uniformly distributed on $\mathbb{S}^1$; for $K = 4.5$ they cluster into a narrow region. Fig. 4 shows the effective frequencies $\Omega_i = \langle\dot{\theta}_i\rangle_t$ as a function of the natural frequencies ω_i. For $K > K_c$ the central plateau predicted by theory appears [2]: oscillators with $|\omega_i| < K r_{\text{est}}$ become locked and rotate at the same frequency, while those with $|\omega_i| > K r_{\text{est}}$ remain free, following approximately $\Omega_i \approx \omega_i$.

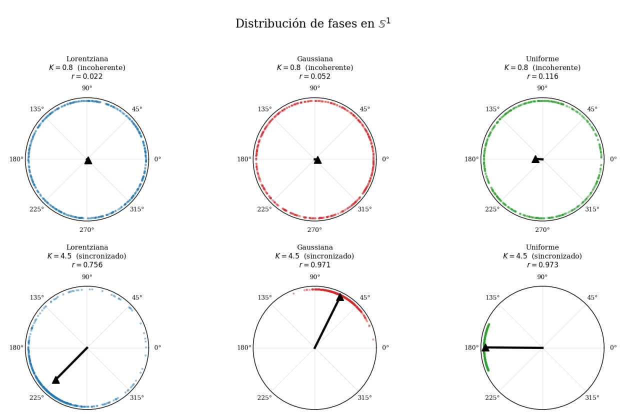

FIG. 3. Snapshots of the phase distribution on $\mathbb{S}^1$ for $K = 0.8$ (top row) and $K = 4.5$ (bottom row), for the three distributions. The black arrow indicates the order-parameter vector.

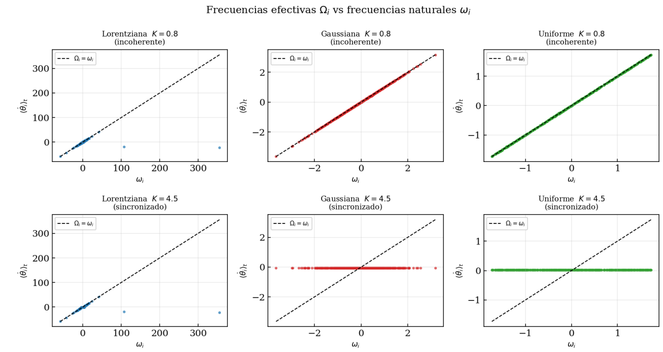

FIG. 4. Effective frequencies Ω_i versus natural frequencies ω_i for the three distribution types, in the incoherent ($K = 0.8$, top row) and synchronized ($K = 4.5$, bottom row) regimes. The central plateau identifies the locked oscillators.

Power spectrum of $r(t)$. Fig. 5 shows the power spectrum of the fluctuations $r(t) - \langle r\rangle$ for three values of K. For $K < K_c$, the spectrum is broad and lacks a dominant peak, consistent with random finite-size fluctuations. For $K \approx K_c$, a spectral broadening is observed, typical of the increased correlation times near a continuous transition. For $K > K_c$, the spectrum concentrates at low frequencies, indicating slow, coherent collective oscillations around the steady-state value of r.

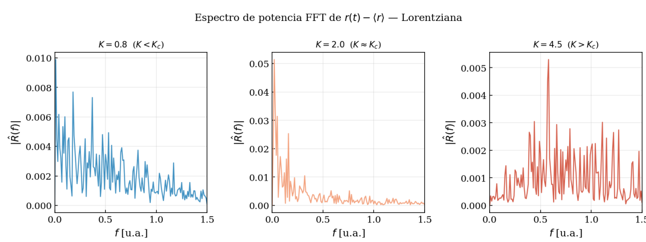

FIG. 5. Power spectrum $|\hat{R}(f)|$ of the fluctuations of $r(t)$ for the three dynamical regimes (Lorentzian distribution). For $K > K_c$ the spectrum concentrates at low frequencies.

Numerical validation. Fig. 6 compares the numerically obtained curve $\langle r\rangle(K)$ for the Lorentzian distribution with the analytical solution $r_{\text{anal}}(K) = \sqrt{1 - K_c/K}$. The agreement is excellent: the root-mean-square error for $K > K_c$ is $\varepsilon_{\text{RMS}} = 0.0167$. The residuals, shown in the bottom panel, are small and display no systematic trend, confirming that the remaining discrepancies are consistent with finite-size effects ($N = 500$ versus the theoretical limit $N \to \infty$).

Collective dynamics and neural analogy. Fig. 7 shows the synthetic collective signal $V_{\text{col}}(t) = (1/N)\sum_j \cos\theta_j(t)$, the evolution of $r(t)$, and the power spectral density (PSD) computed with Welch's method, for three regimes that correspond qualitatively to brain states. For $K = 0.8$, the signal has low amplitude and a broad spectrum, analogous to the cortical resting state

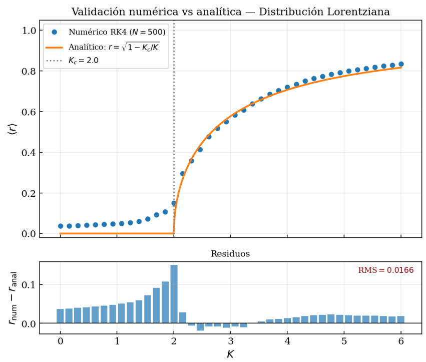

FIG. 6. Comparison between the numerical result (circles) and the analytical solution $r = \sqrt{1 - K_c/K}$ (orange line) for the Lorentzian distribution. The bottom panel shows the point-by-point residuals; the root-mean-square error is $\varepsilon_{\mathrm{RMS}} = 0.0167$.

(α band). For $K = 2.0$, the amplitude grows with large fluctuations, similar to states of cognitive transition (β band). For $K = 4.5$, the signal has large amplitude and is nearly periodic, with a dominant spectral peak analogous to an epileptiform discharge (hypersynchronization).

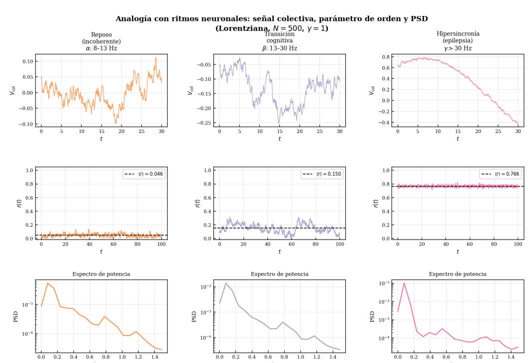

FIG. 7. Neural analogy: collective signal $V_{\mathrm{col}}(t)$ (top row), order parameter $r(t)$ (middle row), and power spectrum PSD (bottom row) for the three dynamical regimes (Lorentzian distribution, $N = 500$). The states correspond qualitatively to cortical rest ($K = 0.8$), cognitive transition ($K = 2.0$), and epileptic hypersynchronization ($K = 4.5$).

DISCUSSION

Effect of the frequency distribution. The results confirm that the functional form of $g(\omega)$ quantitatively determines the characteristics of the transition. The difference between distributions resides mainly in the tails: the Lorentzian has algebraic tails (decay $\sim \omega^{-2}$) that contribute oscillators with frequencies very far from the mean. These extreme oscillators are the last to become locked as K increases, slowing down the saturation of r toward 1 for $K \gg K_c$. The uniform distribution, having compact support, guarantees that all oscillators have frequencies within a bounded interval, which translates into a sharper transition. The Gaussian occupies an intermediate position with its exponential decay.

Finite-size effects. For $N = 500$, the system differs from the theoretical limit $N \to \infty$ in predictable ways. The fluctuation "floor" $\langle r \rangle \sim N^{-1/2} \approx 0.045$ for $K < K_c$ is clearly visible in Fig. 1, as is the smooth rounding of the transition near K_c. These effects are intrinsic to the finite system and do not represent numerical errors: they are the statistical manifestation of the fact that, with finite N, there are always fluctuations around the mean state, analogous to magnetic fluctuations in finite-size ferromagnets [2].

Spectral behavior near K_c. The spectral broadening of $r(t)$ observed in Fig. 5 for $K \approx K_c$ is consistent with the increase in correlation time characteristic of a system near a continuous transition. In the language of phase transitions, this is called *critical slowing down*: the system takes increasingly longer to relax toward equilibrium as it approaches the critical point. This phenomenon has been observed in EEG recordings prior to epileptic seizures, where the brain shows signs of "critical deceleration" before the discharge [4].

The neural analogy: what it captures and what it does not. The connection between the Kuramoto model and neuroscience deserves a careful discussion of its limits. What the model correctly captures is the *mean-field physics* of synchronization: the emergence of collective coordination from heterogeneous coupled oscillators, the existence of a critical threshold, and the coexistence of locked and free populations that determines the value of r. These ingredients are qualitatively correct for describing neuronal populations when synaptic coupling is weak [7].

What the model does *not* capture is the biophysical richness of the real neuron: action potentials with threshold and refractoriness, diverse ionic currents, synaptic plasticity, cortical network architecture (layers, columns), and the distinction between excitatory and inhibitory neurons. A more faithful description requires models such as Hodgkin-Huxley or integrate-and-fire [5]. In this sense, the Kuramoto model is to the brain what the mean-field Ising chain is to a real ferromagnet: it captures the universal physics of the transition but not the material's details.

Nevertheless, this "simplicity with substance" is precisely what makes the model valuable. The hypothesis that epilepsy corresponds to a transition from the normal processing state ($r < 1$) toward hypersynchronization ($r \to 1$) has motivated deep brain stimulation strategies designed to "destabilize" pathological synchronization, which conceptually amounts to reducing the effective K below K_c [4]. This connection between a theoretical physics model and a real clinical application illustrates the explanatory power of mean-field models in biophysics.

CONCLUSIONS

We have carried out a systematic numerical study of the synchronization transition in the Kuramoto model for $N = 500$ oscillators, comparing the Lorentzian, Gaussian, and uniform natural-frequency distributions. The main results are as follows.

First, the transition from incoherent to synchronized is observed in all three cases, confirming the working hypothesis: both the coupling strength K and the shape of $g(\omega)$ determine the characteristics of the transition. The critical coupling is smaller for the Gaussian than for the Lorentzian and uniform distributions, as a consequence of the different tail structure of each distribution.

Second, validation of the RK4 integrator against the exact analytical solution of the Lorentzian case yielded an error $\varepsilon_{\mathrm{RMS}} = 0.0167$, consistent with finite-size effects, confirming the correct implementation of the numerical method and the $\mathcal{O}(N)$ optimization of the derivative computation.

Third, the spectral and effective-frequency analysis numerically confirmed the theoretically predicted coexistence of locked and drifting oscillators for $K > K_c$, as well as the spectral broadening characteristic of *critical slowing down* near K_c.

Fourth, the qualitative analogy with neural rhythms shows that the Kuramoto model is able to reproduce, in mean-field terms, the spectrum of brain states: from normal cortical processing (partial incoherence) to epileptic hypersynchronization ($r \to 1$). This connection not only enriches the physical context of the work but also illustrates how simple mathematical models can offer valuable insights into complex biological systems.

As natural extensions, we propose studying the effect of the number of oscillators N on the transition (finite-size analysis), incorporating stochastic noise into Eq. (1) (stochastic Kuramoto model), and exploring more realistic network topologies (small-world or scale-free networks) as a step toward more elaborate neurocomputational models.

* andres.arrietaand@unipamplona.edu.co

[1] Y. Kuramoto, *Chemical Oscillations, Waves, and Turbulence* (Springer, Berlin, 1984).
[2] J. A. Acebrón, L. L. Bonilla, C. J. Pérez Vicente, F. Ritort, and R. Spigler, Rev. Mod. Phys. **77**, 137 (2005).
[3] S. H. Strogatz, Physica D **143**, 1 (2000).
[4] M. Breakspear, S. Heitmann, and A. Daffertshofer, Front. Hum. Neurosci. **4**, 190 (2010).
[5] D. Cumin and C. P. Unsworth, Physica D **226**, 181 (2007).
[6] G. Buzsáki and A. Draguhn, Science **304**, 1926 (2004).
[7] G. B. Ermentrout and D. H. Terman, *Mathematical Foundations of Neuroscience* (Springer, New York, 2010).
[8] J. Buck, Q. Rev. Biol. **63**, 265 (1988).
[9] F. Dörfler and F. Bullo, Automatica **50**, 1539 (2014).
[10] W. H. Press, S. A. Teukolsky, W. T. Vetterling, and B. P. Flannery, *Numerical Recipes*, 3rd ed. (Cambridge University Press, Cambridge, 2007).